

DEEPANWITA BANERJEE

Computational Biology | Systems Biology | Model & Multi-Omics Driven Microbial Engineering

dipsiii@gmail.com • [linkedin.com/in/deepanwita](https://www.linkedin.com/in/deepanwita) • [Google Scholar](https://scholar.google.com/citations?user=...) • [GitHub](https://github.com/deepanwita)

RESEARCH PROFILE

Research Scientist with 8+ years of experience integrating computational modeling, multi-omics, and experimental microbial engineering. At Berkeley Lab, specialized in *Pseudomonas putida* strain design through model-guided experimental validation and quantitative systems biology. Expert in linking genome-scale models with high-dimensional biological data to drive sustainable carbon utilization and engineer complex microbial systems.

SELECTED PUBLICATIONS

Google Scholar: 970 citations • **h-index:** 16

- **Banerjee, D.**, Yunus, I.S., Wang, X., Kim, J., et al. *Genome-scale and pathway engineering for the sustainable aviation fuel precursor isoprenol production in Pseudomonas putida*. **Metabolic Engineering** 82, 157–170 (2024). [Link](#)
- **Banerjee, D.**, Menasalvas, J., Chen, Y., Gin, J.W., Baidoo, E.E.K., Petzold, C.J., et al. *Addressing genome scale design tradeoffs in Pseudomonas putida for bioconversion of an aromatic carbon source*. **npj Systems Biology and Applications** 11, 8 (2025). [Link](#)
- **Eng, T., Banerjee, D.**, Menasalvas, J., et al. *Maximizing microbial bioproduction from sustainable carbon sources using iterative systems engineering*. **Cell Reports** 42(9) (2023). [Link](#)
- **Banerjee, D.**, Eng, T., Lau, A.K., et al. *Genome-scale metabolic rewiring improves titers, rates and yields of the non-native product indigoidine at scale*. **Nature Communications** 11, 5385 (2020). [Link](#)
- Baral, N.R., **Banerjee, D.**, Eng, T., et al. *Microbial Pathways for Cost-Effective Low-Carbon Renewable Indigoidine*. **ACS Sustain Chem Eng**. 2025 Feb 14;13(8):3300-3310. [Link](#)
- Kulakowski, S., **Banerjee, D.**, Scown, C.D., Mukhopadhyay, A. *Improving microbial bioproduction under low-oxygen conditions*. **Current Opinion in Biotechnology** 84, 103016 (2023). [Link](#)
- **Banerjee, D.**, Mukhopadhyay, A. *Perspectives in growth production trade-off in microbial bioproduction*. **RSC Sustainability** 1(2), 224–233 (2023). [Link](#)
- Bhattacharya, N., **Banerjee, D.**, O'Connor, C., Parmar, D., Moskovets, E., et al. *Direct Colony Atmospheric Pressure Matrix-Assisted Laser Desorption/Ionization Mass Spectrometry Assay for Targeted Operon Metabolite Analysis*. In **Metabolomics: Methods and Protocols** (2025). [Link](#)

PATENTS

- Genetically modified *Pseudomonas* host cells and methods useful for producing isoprenol. US Patent Application 18/976,085 (2025).
- Genetically modified bacterial cells and methods useful for producing indigoidine. US Patent 11,767,521 (2023).

SELECTED PRESENTATIONS

- Engineering *Pseudomonas putida* for efficient conversion of lignin-derived sugars and aromatics into bioproducts and biofuels. SIMB 71st Annual Meeting, Austin, TX, August 2021.
- Genome-scale Metabolic Rewiring for Engineering Robust Production Platforms. Indo-US Workshop on Recent Developments in Bioenergy Research, IIT Kharagpur, October 2020.
- Strain Engineering in *Pseudomonas putida* Using High Throughput Fitness Profiling and Genome Scale Metabolic Models. Metabolic Engineering 14 Conference, July 2021.
- Integrated *Pseudomonas putida* Strain Design for Maximizing Biomass Conversion to Biofuels and Bioproducts. US DOE BER Genomic Sciences Program Annual PI Meeting, February 2021.
- Systems Metabolic Engineering of *Pseudomonas putida* for Indigoidine Production. JBEI Annual DOE Review, December 2019.

PROFESSIONAL EXPERIENCE

Berkeley Lab, BSE Division — Research Scientist, Mukhopadhyay Group | Aug 2021–July 2026

- Integrated metabolic-model predictions with experimental phenotypes to identify metabolic and physiological determinants of microbial behavior.
- Analyzed microbial responses across cultivation modes and environmental regimes to identify system-level constraints and opportunities for strain engineering.
- Investigated microbial physiology under low-oxygen conditions, sustainable carbon utilization, and lignin-derived feedstocks to understand growth–production tradeoffs and metabolic adaptation.
- Contributed to iterative computational–experimental workflows for identifying and validating strategies to improve microbial production.

Joint BioEnergy Institute (JBEI) / Berkeley Lab — Postdoctoral Fellow, Mukhopadhyay Group | Oct 2018–Aug 2021

- Developed a scalable model-integrated computational workflow for strain design and prediction of engineering strategies in *Pseudomonas putida* KT2440.
- Applied systems metabolic engineering to connect pathway-level perturbations, cellular physiology, and strain-level production performance.
- Used high-throughput fitness profiling and genome-scale metabolic models to investigate *P. putida* strain behavior and identify candidate engineering strategies.
- Integrated computational predictions with experimental strain construction and phenotypic validation.
- Presented research on model-guided *P. putida* engineering and genome-scale metabolic modeling at national and international meetings.

CSIR–National Chemical Laboratory — DST INSPIRE PhD Fellow | Aug 2011–May 2018

- Built and applied constraint-based metabolic models using COBRA Toolbox/MATLAB to investigate microbial metabolism, redox homeostasis, and antibiotic resistance.
- Combined computational and experimental approaches to investigate microbial metabolic adaptation.

MENTORING & LEADERSHIP

- Mentored SULI interns on genome-scale metabolic modeling and computational strain-design projects at JBEI.
- Supervised undergraduate and master's research students and managed laboratory interns during Ph.D.
- **Awards** - 2021-JBEI Research Contribution, 2020-Berkeley Lab SPOT Recognition, 2020-JBEI Outreach & Community Contribution

CORE EXPERTISE

Computational & Systems Biology - Genome-scale metabolic reconstruction, modeling & Analysis • Model-guided hypothesis generation • MATLAB • Jupyter Notebook • GitHub • COBRA Toolbox • Elementary Mode Analysis

Microbial Systems & Multi-Omics - *Pseudomonas putida* • Multi-omics analysis • Growth & production phenotyping • High-throughput -omics profiling • Microbial response to environment • Metabolic and physiological heterogeneity

Experimental Microbial Engineering - Cloning • PCR • Lambda-red gene knockout • BIOLOG • Basic Biochemical & microbiological assays

Metabolic Engineering & Bioproduction - Genome-scale strain engineering • Growth–production tradeoffs • Sustainable bioproduction • Carbon utilization

EDUCATION

- Ph.D., Biological Sciences — CSIR–National Chemical Laboratory, Pune, India | 2018
- M.Tech., Biotechnology — National Institute of Technology Rourkela, India | 2009
- Institute Silver Medal • First Rank
- B.Tech., Biotechnology — Maharshi Dayanand University, India | 2007